

Exercise 1: Control Structures

Scenario 1: Apply Discount to Loan Interest Rates

```
BEGIN

  FOR customer_rec IN (
    SELECT CustomerID
    FROM Customers
    WHERE Age > 60
  )
  LOOP
    UPDATE Loans
    SET InterestRate = InterestRate - 1
    WHERE CustomerID = customer_rec.CustomerID;
  END LOOP;

  COMMIT;

  DBMS_OUTPUT.PUT_LINE('1% discount applied to eligible customers.');
```

END;

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Scenario 2: Promote Customers to VIP Status

```
BEGIN

  FOR customer_rec IN (
    SELECT CustomerID
    FROM Customers
    WHERE Balance > 10000
  )
```

```
LOOP

    UPDATE Customers

    SET IsVIP = 'TRUE'

    WHERE CustomerID = customer_rec.CustomerID;

END LOOP;

COMMIT;

DBMS_OUTPUT.PUT_LINE('VIP status updated successfully.');
```

END;

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Scenario 3: Send Loan Due Reminders

```
SET SERVEROUTPUT ON;

BEGIN

    FOR loan_rec IN (

        SELECT c.Name,

               l.LoanID,

               l.DueDate

        FROM Customers c

        JOIN Loans l

        ON c.CustomerID = l.CustomerID

        WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30

    )

    LOOP

        DBMS_OUTPUT.PUT_LINE(

            'Reminder: Dear ' || loan_rec.Name ||

            ', your Loan ID ' || loan_rec.LoanID ||

            ' is due on ' || TO_CHAR(loan_rec.DueDate, 'DD-MON-YYYY')
```

);

END LOOP;

END;

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Exercise 3: Stored Procedures

Scenario 1: Process Monthly Interest

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
AS
BEGIN
    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'SAVINGS';

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Monthly interest processed successfully.');
```

END;

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Scenario 2: Update Employee Bonus

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (
    p_Department IN VARCHAR2,
    p_BonusPercent IN NUMBER
)
AS
BEGIN
    UPDATE Employees
    SET Salary = Salary + (Salary * p_BonusPercent / 100)
    WHERE Department = p_Department;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Employee bonus updated successfully.');
```

END;

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Scenario 3: Transfer Funds Between Accounts

```
CREATE OR REPLACE PROCEDURE TransferFunds (  
    p_FromAccount IN NUMBER,  
    p_ToAccount IN NUMBER,  
    p_Amount IN NUMBER  
)  
AS  
    v_Balance NUMBER;  
BEGIN  
    SELECT Balance  
    INTO v_Balance  
    FROM Accounts  
    WHERE AccountID = p_FromAccount;  
  
    IF v_Balance >= p_Amount THEN  
  
        UPDATE Accounts  
        SET Balance = Balance - p_Amount  
        WHERE AccountID = p_FromAccount;  
  
        UPDATE Accounts  
        SET Balance = Balance + p_Amount  
        WHERE AccountID = p_ToAccount;  
  
        COMMIT;  
  
        DBMS_OUTPUT.PUT_LINE('Funds transferred successfully.');
```

ELSE

DBMS_OUTPUT.PUT_LINE('Insufficient balance.');

END IF;

END;

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